

SELECT with various clause – BETWEEN, IN, Aggregate function

AIM:

To view the records from the tables using SELECT commands with BETWEEN, IN, Aggregate

functions.

IN & BETWEEN

1. List the type of the courses "Statistics" and "Programming"

```
mysql> SELECT CourseNo
-> FROM REGISTER
-> WHERE Score BETWEEN 50 AND 80;
+-----+
| CourseNo |
+-----+
| C001      |
| C002      |
| C003      |
+-----+
3 rows in set (0.04 sec)

mysql> SELECT CourseType
-> FROM Course
-> WHERE CourseDescription IN ('Statistics','Programming');
+-----+
| CourseType |
+-----+
| Theory     |
| Practical  |
+-----+
2 rows in set (0.00 sec)
```

2. Display the CourseNos whose scores are between 50 and 80

```
mysql> SELECT CourseNo
-> FROM Register
-> WHERE Score BETWEEN 50 AND 80;
+-----+
| CourseNo |
+-----+
| C001      |
| C002      |
| C003      |
+-----+
3 rows in set (0.00 sec)
```

AGGREGATE FUNCTIONS

1. Find the average mark of "C002"

```

1. Find the average mark of "C002"
SELECT AVG(Score) AS Aver' at line 1
mysql> SELECT AVG(Score) AS Average_Mark
      -> FROM Register
      -> WHERE CourseNo='C002';
+-----+
| Average_Mark |
+-----+
|      85.0000 |
+-----+
1 row in set (0.01 sec)

mysql> |

```

2. List the maximum and minimum mark for "C002"

```

mysql> SELECT MAX(Score) AS Maximum_Mark,
      -> MIN(Score) AS Minimum_Mark
      -> FROM Register
      -> WHERE CourseNo='C002';
+-----+-----+
| Maximum_Mark | Minimum_Mark |
+-----+-----+
|           95 |           75 |
+-----+-----+
1 row in set (0.01 sec)

mysql> |

```

3. List the maximum, minimum and average mark for each course

```

mysql> SELECT CourseNo,
      -> MAX(Score) AS Maximum_Mark,
      -> MIN(Score) AS Minimum_Mark,
      -> AVG(Score) AS Average_Mark
      -> FROM Register
      -> GROUP BY CourseNo;
+-----+-----+-----+-----+
| CourseNo | Maximum_Mark | Minimum_Mark | Average_Mark |
+-----+-----+-----+-----+
| C001     |           65 |           45 |      55.0000 |
| C002     |           95 |           75 |      85.0000 |
| C003     |           55 |           55 |      55.0000 |
+-----+-----+-----+-----+
3 rows in set (0.04 sec)

```

4. List the course name and average mark of each course

```
mysql> SELECT C.CourseDescription,
->         AVG(R.Score) AS Average_Mark
-> FROM Course C
-> JOIN Register R
-> ON C.CourseNo = R.CourseNo
-> GROUP BY C.CourseDescription;
```

CourseDescription	Average_Mark
Statistics	55.0000
Programming	85.0000
Database	55.0000

3 rows in set (0.01 sec)

5. Calculate the sum of all the scores

```
mysql> SELECT SUM(Score) AS Total_Score
-> FROM Register;
```

Total_Score
420

1 row in set (0.00 sec)

6. Display the course description and the number of students registered in each course

```
mysql> SELECT C.CourseDescription,
->         COUNT(R.RegNo) AS Number_of_Students
-> FROM Course C
-> JOIN Register R
-> ON C.CourseNo = R.CourseNo
-> GROUP BY C.CourseDescription;
```

CourseDescription	Number_of_Students
Statistics	2
Programming	3
Database	1

3 rows in set (0.00 sec)